

Spillover across varying Regional and Economic Realities: Evaluating the Hedging potential of Demographic Traits

Abstract

Advancements in technology are increasingly creating a highly integrated and connected financial environment across the globe. The resulting financial contagion has also led to increased volatility and uncertainties for the stakeholders. It is becoming increasingly difficult for the market participants to diversify, which has led to an increased focus on the avenue. Therefore, the current investigation utilizes behavioral theory to focus on varying regional and developmental realities across the globe to evaluate their connectedness and hedging potential for the investors. The study utilizes data from ten representatives stock market indices based on MSCI classification of regions and development ranging from 2012 till 2024. The dualistic analysis initiated by utilizing VAR based connectedness measures to evaluate the degree of connectedness among select indices while an appropriate GARCH model based on selection criteria is applied to test for hedging effectiveness. The connectedness analysis reveals an overall moderate degree of connectedness across these stock indices with weak spillover across time elevating significantly during COVID-19 pandemic. The degree of spillovers diminishes across regions and with decreasing level of market development. Meanwhile the hedging analysis revealed that investment across regional and developmental spectrum of stock markets results in achieving significantly elevated portfolio efficiency for investors. The current study is a pioneer to test MSCI classification-based variations to evaluate the degree of their connectedness and hedging ability. The study brought new incremental insights on the dynamics of stock market transmissions from behavioral perspective. Furthermore, the findings of connectedness and hedging have practical insights for the stakeholder for risk diversification and to enhance portfolio returns.

Keywords: MSCI Classification, Stock Markets, Connectedness, Spillover, Hedging

1. Introduction

Stock markets around the globe are detrimental for growth and development of an economy. Since the start of 21st century, an exponential increase in technology has led to a higher degree of final integration and contagion across the global markets. There is an increasing recognition that these factors are leading to higher uncertainty and is making economic growth highly sensitive and unsustainable (Cleary, 2017). Traditionally the stock markets movements are considered to be efficient and assumed to follow a random walk (Bloomfield & Hales, 2002) but a dominant array of evidence indicates otherwise. Based on the traditional theories of financial behaviour, the recent advancements in communication are expected to better equip the market participants to perform rational and optimal

financial choices leading to efficient markets but rise in technology is increasingly making the stock markets volatile and unstable (Rasheed et al., 2023). Even since the pioneering work of Markowitz in (1952), the avenue of stock markets movements and returns always remained an avenue of interest for the investors, researchers, practitioners and policymakers. This heightened financial contagion and innovations has regained attention of the researchers around the globe. Particularly at micro level, the technological innovations like artificial intelligence (AI), blockchain and fintech etc. is constantly expanding the investment horizons. The investors' ability and availability of assets is constantly and exponentially increasing. This increased economic opportunities when coupled with increased global sensitivity is making it difficult for the portfolio holders to effectively gain optimal returns and diversify. Hence, the era of conventional financial standardization across economies is diminishing rapidly and the focus of the researchers has shifted towards developing a contextual understanding of the economic realities across the globe (Rasheed et al., 2024; Q. Tan et al., 2024).

Twenty first century's financial literature is marked with substantial application of technological advancements and ensuring planet sustainability for the future generations. These factors have provided modern researchers with a diverse array of new investment avenues and hence capture a dominant portion of contemporary literature. The inception of blockchain technologies and its acceptance is increasing with a wide spectrum of applications for overall economic environment (Yousaf et al., 2023a). It has been able to receive significant attention from investors, media hype, regulatory bodies across the globe, and academics from different disciplines (Symitsi & Chalvatzis, 2018). Ample literature has on the topic has analysed the connectedness and hedging abilities between cryptocurrencies and other asset classes including green assets (Shao et al., 2023), cryptocurrencies and the stock market (Yen, 2023) cryptocurrency, and cryptocurrency-related stocks (Frankovic et al., 2022), Artificial Intelligence (AI) tokens and AI stocks (Jareño F; Yousaf, 2023), real estate tokens and REITs (Abdullah et al., 2023; Yousaf et al., 2024), meme tokens and meme stocks (Yousaf et al., 2023b), fan tokens and football club stocks (Ersan et al., 2022) to name a few among continuously evolving studies on the blockchain.

Similarly, on the avenue of sustainability Considerable efforts have been made to confront the approaching danger, such as the establishment of the Sustainable Development Goals (SDGs) for 2030 and the inclusion of National Determined Contributions (NDC) in the Paris Agreement. (Ozili, 2021). Which has led to the inception of green finance, financial models that would be created to ensure the sustainability of the earth for future generations (Hunjra et al., 2023). The avenue has contributed modern and environmentally friendly financial concepts and products, such as carbon trading and green bonds (GBs) (Chopra & Mehta, 2023). This demonstrates the potential and influence of green finance in reducing this grave threat to human survival. These new financial instruments have enabled and encouraged researchers to study the connectedness and hedging abilities of green bonds (Broadstock et al., 2022), ESG investments (Liu et al., 2023) and green finance (Wan et al., 2023) to name a few. Whereas, another school of researchers like Jangid and Bhardwaj (2023), Kuang et al. (2024), and Peng et al. (2023) are working on unification of the technological innovations and sustainability. These trending avenues foreshadowed other avenues of financial literature and modern researchers overlooked the investing and hedging potential some very relevant factors. Therefore, this study is aimed at shifting the focus on those avenues and one such avenue is the avenue of behavioural differences across the globe. Thaler (2015) highlighted a need to explore behavioural differences at macro level, therefore the current investigation is an attempt to fill this significant research gap in the field of connectedness and hedging abilities.

The inception of behavioural finance can be attributed to the prospect theory with pioneering work of Kahneman and Tversky (1979). The theory challenges the traditional financial paradigm of uniform rational and efficient behaviour and posits that behaviour across losses and gains vary significantly. Since its inception, it has become a widely accepted model of financial behaviour across the globe. Demographic and cognitive factors are now a well-established determinants of contextualized and unique financial behaviour across the globe (Bhatia A; Chandani A; Chhateja, 2020; Gharbi et al., 2023; Kudryavtsev et al., 2013; Rasheed et al., 2021). Stock market behaviour is found to be vary significantly across ethnicities, cultures, and borders (Q. Tan et al., 2024). The study of Burnham et al. (2018) found that the Morgan Stanley Capital International (MSCI) categorization of stock markets reclassification significantly influence a stock market behaviour. Similarly, the study by Tan et al. (2024) also concluded significant difference across regional developmental classification of stock markets. Following the patterns of Broadstock et al. (2022), a bibliometric shortlisting of the literature based on the Web of Science (WOS) and Scopus corpus is conducted to uncover the recent trends of the MSCI classifications. The inquiry uncovered that the use of MSCI classification across various avenues of financial literature is witnessing an exponential increase in the recent times. This widespread adoption of MSCI indices indicates reliability and validity of their representation for financial literature. Recently researchers like Barandica et al. (2024) studied corporate profitability in Colombia, Bayat et al. (2024) relationship between ETF and FX across Türkiye, Tan and Zhu (2024) studied corporate innovation abilities across China, Balestra et al. (2024) studied influence of armed conflicts on wealth funds, Waris and Din (2024) studied role of government for environmental sustainability in Malaysia and Sharma et al. (2024) studied diversification using cryptocurrencies by relying on MSCI based indices. This indicates the range of avenues and demography covered by MSCI, this high reliance also establishes the classification as a valid criterion for any future inquiry. As far as the avenue of connectedness and hedging is concerned the modern avenue overlooked the potential of considering the classification itself for any diversification. Instead researchers like Chancharat and Sinlapates (2024), Barandica et al. (2024), Sharma et al. (2024), and Nekhili et al. (2024) focused on the comparison of MSCI index with cryptocurrencies, pacific region, fintech, centralization. While only a few of the studies on this topic extended to include the hedging potential of such markets like Muvunza and Jiang (2023), Nugroho and Kusumawardhani (2023), and Ustaoglu (2023). Still, none of these focused on the potential of MSCI regional and developmental categorization for diversification. Based on these findings the current study is aimed at extending the current literature on regional and developmental variation to test for connectedness. By utilizing these behavioural variations, the authors expect to squeeze the hedging potential of the indexed portfolios. The current investigation is a pioneer to consider and study the connectedness and hedging potential of such difference based on behavioural theories and has brought substantial insights for the stakeholders.

2. Methodology

The focus of the study is to explore and establish the differences in stock markets across Morgan Stanley Capital International (MSCI) regional and developmental classification of stock markets around the globe. Contrary to traditional view of uniform behaviour, random walk and market efficiency, these behavioural deviations of the markets create varying degree of spillover and connectedness among each other creating possibilities for portfolio diversification. To establish and validate the investment diversification potential of these market appropriate statistical models backed by reasonable sample selection is mandatory. Therefore, this section focuses on the issue of justifiable sample and methodology selection for valid and reliable inference of the research questions.

Sample Selection

Morgan Stanley Capital International (MSCI) is a research enterprise based in New York, USA. It provides benchmarks and indices for the stock markets around the globe and is by far the most popular and acknowledged for investing across global equity markets (Burnham et al., 2018). It categorises global stock markets based on their maturity and development into developed, developing, frontier and standalone markets, while it also creates a regional distinction among Americas, Europe, Middle East & Africa (EMEA), and Asian Pacific (APAC) markets (Q. Tan et al., 2024). Hence provides a comprehensive and diverse set of markets with varying attributes. Contrary to the traditional view of uniform behaviour, existing literature including work of Burnham et al. (2018) and Singh et al. (2019) observed significant difference across these attributes. This provides an ideal platform to test these variations to uncover the patterns of spillover among them leading to hedging opportunities for the investors globally.

Table 1- MSCI Classification of Markets

Classification	Region	Sample	Index
Developed	America	USA(US)	S&P 500
	EMEA	Germany (DE)	GDAXI
	APAC	Japan (JP)	Nikkei 225
Emerging	America	Brazil (BR)	BOVESPA
	EMEA	Saudi Arabia (SA)	TASI
	APAC	China (CN)	SSCEI
Frontier	America	None	
	EMEA	Bahrain (BH)	BAX
	APAC	Pakistan (PK)	KSE-100
Standalone	America	Jamaica (JM)	JSE
	EMEA	Palestine (PS)	PLE
	APAC	None	

Table 1 represent the stock indices selected for the final sample of the study across each market categorization. These economies are selected based on data availability and the authors utilized Yahoo finance to create final dataset for the study. The final sample comprises of data from 10 stock markets across the globe. The underlying reason for 10 markets in final sample is that MSCI currently does not categorize any frontier market from Americas and neither it categorizes any standalone market from Asian Pacific region. The data for each stock market indices range from 2012 till 2024 providing sufficient observations for a valid and generalizable statistical inference while also covering extreme financial events like COVID-19 pandemic. The data comprises of log returns calculated from the daily closing values of each representative stock market. Figure 1 below presents

the time varying dynamics of each stock market index. It can be observed that the developed markets provide much more stable returns, and the market volatility increases as it moves toward less developed end of the spectrum. This variability of stock returns further affirms the rationale of the study.

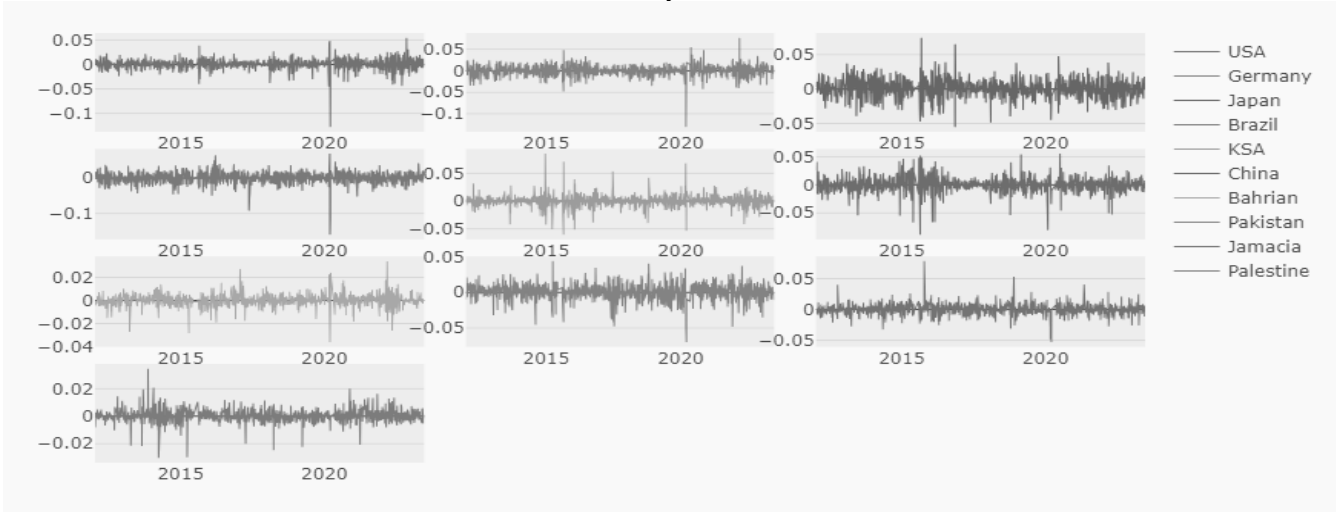


Figure 1- Index Returns Plot

Methodology

The current study focuses on the degree of connectedness and its hedging potential across stock markets with varying economic and regional dynamics. Therefore, the study is motivated based on multifaceted considerations, including spillover across markets and regions, to identify, measure, and monitor spillovers. This eventually enables investors to select diversifying markets or assets based on the interconnectedness and hedging potential. To achieve this objective, the study utilises the spillover model pioneered by Diebold and Yilmaz (2009) and further extended by Diebold and Yilmaz in (2012). They proposed a volatility spillover measure based on the vector autoregressive model (VAR), which enables researchers to analyse spillover within and across assets, asset classes, asset markets, and regions in every possible combination. The final derivations of various statistical outcomes of the model are reported in Table 2 below.

Table 2- Spillover Model

1-Variance share of each asset	2-Total average spillover
$\tilde{\theta}_{ij}^g(H) = \frac{\theta_{ij}^g(H)}{\sum_{j=1}^N \theta_{ij}^g(H)}$	$TCI^g(H) = \frac{\sum_{i,j=1}^N \tilde{\theta}_{ij}^g(H)}{N} \cdot 100$
3- Spillover to an asset	4-Spillover from an asset
$TO_i^g(H) = \frac{\sum_{j=1}^N \tilde{\theta}_{ij}^g(H)}{N} \cdot 100$	$FROM_i^g(H) = \frac{\sum_{j=1}^N \tilde{\theta}_{ji}^g(H)}{N} \cdot 100$
5-Net average spillover from an asset	6-Net pairwise spillover from an asset
$NET_i^g(H) = FROM_i^g(H) - TO_i^g(H)$	$PAIR_{ij}^g(H) = \frac{(\tilde{\theta}_{ji}^g(H) - \tilde{\theta}_{ij}^g(H))}{N} \cdot 100$

Furthermore, the study also aimed to extend, validate and apply the descriptives of the spillover model for hedging purposes. To test for the dynamics of the hedging potential of these distinct markets, the study is utilising the method utilised by Broadstock et al. (2022). The model will establish the financial implication of these findings for the stakeholders. Overall, the model relies on the study of Antonakakis et al. (2021) For portfolio constructions. The study utilised the Markowitz (1952) based variance and Christoffersen et al. (2014) Based correlational portfolios. Finally, to gauge the hedging potential of the portfolio, the study utilised Sharpe (1966) ratio, hedge ratio based on Kroner and Sultan

(1993) and hedge effectiveness (HE) measure by Ederington (1979). The final derivation of the hedging process and measures utilised in the study are reported in Table 3 below.

Table 3- Hedging Measures

1-Minimum Variance Weights (VW)	2- Minimum Correlation Weights (CW)
$VW_t = \frac{H_T^{-1}I}{IH_T^{-1}I}$	$CW_t = \frac{R_T^{-1}I}{IR_T^{-1}I}$
3- Conditional Correlation (R)	4-Sharpe Ratio (SR)
$R = diag(H_t)^{-0.5} - H_t diag(H_t)^{-0.5}$	$SR = \frac{\bar{r}_p}{\sqrt{Var(r_p)}}$
5- Hedge Ratio (HR)	6-Hedging Effectiveness (HE)
$HR_{ij,t} = \frac{CoVaR_{ij,t}}{Var_{jj,t}}$	$HE = 1 - \frac{Var(y_p)}{Var(y_{unhedged})}$

The models and their measures are widely utilised in spillover, connectedness and hedging. The recent rise of technology leading to increased risks and volatilities have led the modern researchers including Broadstock et al. (2022), Hoque et al. (2023), Khan et al. (2023), Mensi et al. (2018), and Yousaf et al. (2023) to study these previously unexplored avenues of connectedness for potential implications for the stakeholders. Hence, the models are widely accepted and robust for studying the spatial dynamics of financial series across various variables and avenues.

3. Results

The methodology section of the study provides grounds for a valid scientific assessment of the underlying data. Appropriate connectedness and hedging models ensure that the descriptive and inferential statistics findings are valid and generalisable. This section aims to evaluate and support the proposition of the current model. For this purpose, the study utilised RStudio and EViews for final analysis, where the utilised VAR model relied on the Akaike Information Criterion (AIC) as a lag selection criterion while considering a rolling window size of 200 in the final analysis, and the hedging model relied on selection criteria for appropriate GARCH model selection. The resulting inferences are reported in the subsequent sections.

3.1. Descriptive Statistics

Table 4 presents the descriptive statistics for the sample data. As discussed earlier, the final dataset comprises stock market indices from 10 economies across the globe. The dated series for each economy comprises daily stock index returns. The function reporting descriptive statistics is twofold: first, it provides an overview of the whole data, and second, it highlights the presence of any significant outliers in the data. The findings for the current dataset indicate that the return and risk (standard deviation) associated with developed and developing markets is higher than those of frontier and standalone markets. These findings establish the widespread nature of the behavioural variations across global economies and validate the current study's rationale. The findings of Jarque Bera (J-B) and augmented Dickey-Fuller (ADF) tests are statistically significant at ($p < 0.01$) and thereby highlighted through bold characters. The findings of these tests failed to reject the alternative hypothesis. The results of the Jarque Bera (J-B) test indicate that the return series doesn't follow a normal distribution. The issue of non-normality is inherent to the financial time series, and the methodology applied in the current investigation addresses such concerns. Similarly, the log transformation of the return series is expected to create stationarity in the time series for final analysis, which is established through statistically significant values of the augmented Dickey-Fuller (ADF) test.

Table 4- Descriptive Summary

	US	DE	JP	BR	SA	CN	BH	PK	JM	PS
Mean	0.0004	0.0005	0.0005	0.0001	0.0004	0.0001	0.0001	0.0003	0.0002	0.0003
Median	0.0006	0.0008	0.0007	0.0003	0.0009	0.0003	0.0000	0.0004	0.0002	0.0002
Max	0.0540	0.0762	0.0743	0.0690	0.0855	0.0555	0.0342	0.0442	0.0787	0.0345
Min	-0.1277	-0.1305	-0.0551	-0.1599	-0.0606	-0.0887	-0.0362	-0.0699	-0.0531	-0.0302
Std.Dev.	0.0104	0.0119	0.0122	0.0155	0.0100	0.0126	0.0049	0.0105	0.0084	0.0044
Skew	-2.0696	-0.9470	-0.0576	-1.7094	0.1427	-0.7816	-0.3626	-0.7212	0.3705	-0.1288
Kurt	25.6096	15.1314	5.5987	18.6745	13.1617	10.3579	10.5893	7.4312	12.9047	11.4664
J-B	33878.8	09667.4	00433.9	16504.5	6626.8	03628.4	03727.2	01392.6	06326.1	04600.8
ADF	-36.708	-39.717	-41.676	-35.355	-38.725	-38.538	-35.084	-36.325	-38.126	-34.188

3.2. Correlational Analysis

The correlational analysis aims to study the degree of co-movement of the stock market indices included in the sample. The findings of the correlational analysis are reported in Table 5 below. The authors have highlighted the top 3 strongest and weakest degrees of association among these markets. It can be observed from the findings that the inter- and intra-return association between developed and developing economies is significantly higher than the association of frontier and standalone markets. Such a significant and wide array of variation in stock indices also validates the authors' initial proposition regarding behavioural variation as a substantial global determinant of varying stock market behaviours. Overall, the findings of the correlational analysis revealed that the higher the market development, the higher its association with other global markets.

Table 4- Correlational Analysis

	US	DE	JP	BR	SA	CN	BH	PK	JM	PS
US	1.000									
DE	0.564	1.000								
JP	0.168	0.252	1.000							
BR	0.523	0.366	0.141	1.000						
SA	0.198	0.263	0.194	0.190	1.000					
CN	0.128	0.130	0.280	0.136	0.102	1.000				
BH	0.108	0.096	0.088	0.128	0.182	0.045	1.000			
PK	0.147	0.155	0.156	0.125	0.133	0.116	0.057	1.000		
JM	0.057	0.038	0.014	0.067	-0.020	0.016	0.013	0.038	1.000	
PS	0.084	0.057	0.072	0.021	0.055	0.066	0.023	0.049	0.015	1.000

Connectedness Analysis

The descriptive and correlational analysis findings provide enough evidence to consider the dataset for VAR-based connectedness analysis. Hence, the connectedness through spillover analysis findings is reported in Table 6. The findings indicate a moderate level of connectedness among selected stock market indices with a total connected index (TCI) of 26.54% and a corrected total connectedness index. The total connectedness score from and towards markets in the sample is 265.38. These findings are aligned with the existing literature on connectedness, where moderate association is found to reduce portfolio risks significantly (Broadstock et al., 2022; Diebold & Yilmaz, 2009; Hoque et al., 2023) Table 6 reports the detailed findings of the analysis. The developed, developing,

frontier, and standalone markets are highlighted in varying colours to help readers understand the deviations, whereas regional differences are highlighted through bold italics and simple writing styles for the Americas, Europe, Middle East & Africa (EMEA), and Asian Pacific (APAC), respectively.

The overall findings established the United States, Germany and Brazil as net transmitters of stock market spillover with a score of 20.66, 08.16 and 07.44, respectively. In contrast, the remaining markets are net receivers of the stock market spillover. Hence, it can be inferred that the global Western markets of the Americas and Europe are the origin of spillover for the international financial system. In contrast, the Middle East, Africa and Asia Pacific remain at the receiving end of the economic system spillover.

Table 6- Spillover Findings

	US	DE	JP	BR	SA	CN	BH	PK	JM	PS
US	57.50	<i>17.87</i>	02.70	12.76	<i>02.79</i>	01.37	<i>01.82</i>	01.20	00.92	<i>01.03</i>
DE	19.40	<i>58.54</i>	04.36	07.37	<i>04.01</i>	01.66	<i>01.44</i>	01.29	01.08	<i>00.85</i>
JP	10.10	<i>09.35</i>	63.36	03.63	<i>02.93</i>	05.54	<i>01.91</i>	01.88	00.63	<i>00.72</i>
BR	14.30	<i>07.09</i>	01.82	67.37	<i>02.53</i>	01.58	<i>01.77</i>	01.50	01.08	<i>00.94</i>
SA	04.50	<i>04.66</i>	02.84	03.67	<i>74.15</i>	01.87	<i>03.68</i>	02.30	00.94	<i>01.39</i>
CN	03.60	<i>03.01</i>	06.61	02.44	<i>02.11</i>	77.47	<i>00.97</i>	01.83	01.15	<i>00.77</i>
BH	02.50	<i>01.73</i>	01.56	02.53	<i>04.13</i>	01.40	<i>82.74</i>	01.19	01.00	<i>01.21</i>
PK	04.60	<i>02.59</i>	01.83	03.49	<i>02.46</i>	01.76	<i>01.49</i>	80.41	00.50	<i>00.81</i>
JM	02.10	<i>02.23</i>	01.45	02.83	<i>02.02</i>	01.34	<i>01.15</i>	00.75	85.16	<i>00.95</i>
PS	01.90	<i>01.11</i>	01.12	01.37	<i>02.43</i>	01.05	<i>01.56</i>	00.77	00.79	<i>87.88</i>
TO	63.10	49.62	24.29	40.07	25.43	17.58	15.79	12.70	08.09	08.68
Own	120.6	108.2	087.7	107.4	099.6	095.1	098.5	093.1	093.3	096.6
FROM	42.40	41.40	36.60	32.60	25.80	22.50	17.30	19.50	14.80	12.10
NET	20.66	08.16	-12.35	07.44	-00.42	-04.95	-01.47	-06.89	-06.75	-03.44

Now, focusing on the individual findings for each market index, the first evident revelation from the finding is that the primary contributor of the stock index movement for each market is their shocks as represented by the diagonal values in Table 6. Furthermore, this attribution tends to increase significantly as the values move towards less developed economies and markets. These findings are consistent with the existing literature on connectedness and spillover by Ali et al. (2023) and Yousaf et al. (2023a). Furthermore, the study also affirms the findings of Broadstock et al. (2022) that the American and European markets are more tightly connected than their Asian Pacific counterparts. They further identified that the problem with these average results is that sometimes, these findings mask time-varying realities associated with the dataset. Therefore, the current study reported in Figure 2 the cumulative time-varying dynamics of the association between these selected market indices. A deeper exploration of the stock market connectedness reveals that the connectedness remains weakly moderate, between 20 to 30%, during the sampled period. Still, the underlying dynamics observed an exponential rise between 2019 and 2022, reaching above 50%. This underlying period corresponds to the global COVID-19 pandemic, and the findings are consistent with the earlier literature of Baker et al. (2020), Broadstock et al. (2022) and Yousaf et al. (2023a). These studies reported a significantly higher degree of association during financial crises and extreme conditions. Like its start, the elevated levels of connectedness normalised in the post-pandemic era across the sampled market indices.

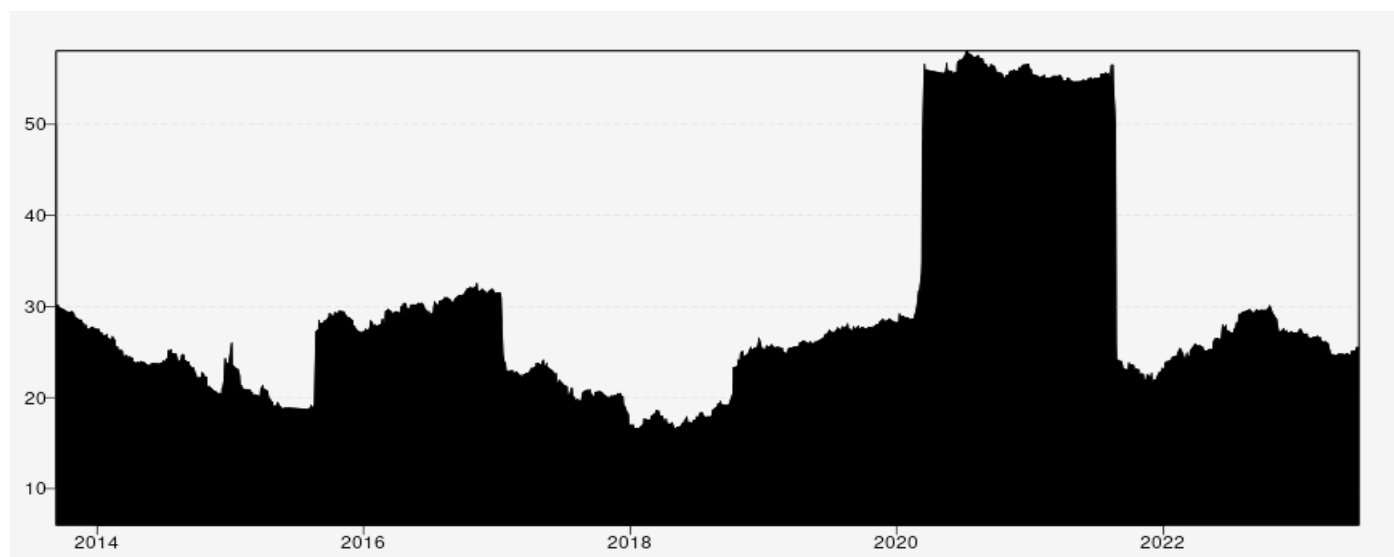


Figure 2-Dynamic Total Connectedness

This has brought us to the dynamics of individual stock markets as recipients and contributors of spillover among each other. The results in Table 6 provide a comprehensive summary of these statistics. It can be observed that spillover from USA to Germany, Germany to USA, and USA to Brazil is the highest at 19.4%, 17.87%, and 14.33%, respectively. In contrast, the spillover from Palestine to Japan, Jamaica to Japan, and Jamaica to Pakistan is the lowest at 0.72, 0.63, and 0.50, respectively. These findings concur with the study of Broadstock et al. (2022) and Tan et al. (2024) regarding significant variations in stock index behaviour based on regions and degree of development. These findings establish that stock market connectedness is significantly higher among American and European markets. The Asian Pacific region has the least spillover across these markets alongside Middle Eastern and African markets. Further analysis of market development reveals that the market connectedness diminishes as it moves towards less developed markets. Lastly, regarding the spillover from each market index, the finding here also uncovers a consistent pattern, where the index associated with the USA, Germany and Japan are the significant spillover transmitters across the globe. In contrast, the standalone markets of Jamaica and Palestine are found to have negligible transmission to other global markets. Hence, a statistical validation affirming the study's rationale is obtained for further analysis.

A spatial representation of varying net connectedness is presented in Figure 3 to gauge the time-varying dynamics of individual stock market indices. It can be observed from the findings that the USA consistently remained the net contributors towards market spillover in the sample, which became significantly higher during the pandemic, while Germany also influenced stock markets throughout the sampled period with substantially higher spillover during and post-pandemic period. At the same time, markets from the Asian Pacific region, including Japan, China, and Pakistan, remain at the receiving end of the spillover. For the Japanese market, the results are consistent (10%-20%) throughout the sampled period, while for China and Pakistan, the reception of weak spillover elevated significantly during the pandemic. The trend of more substantial spillover among developed and developing economies persisted, except for the period of the pandemic. During the pandemic, the role of recipients and contributors intensified but remained uniform except for Brazil, which became a significant transmitter of spillover across the globe. Overall, this persistence of direction, even during times of financial crisis and uncertainties, further highlights the ability of behavioural factors to hedge against increased volatilities and risks.

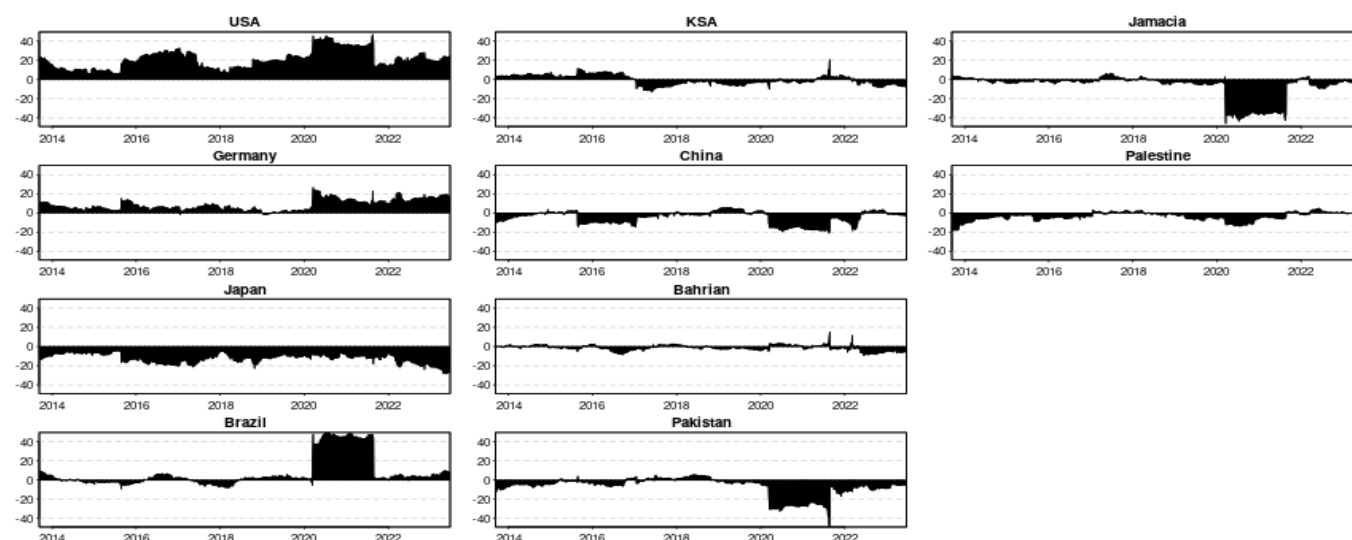


Figure 3-Net Total Connectedness

Even though the average and time-varying analysis of the spillover from stock market indices provided relevant and valuable insights, to delve deeper into the time-varying dynamics of sampled indices, a pairwise representation of the spillover dynamics is reported in Figure 4. A clear pattern can also be observed against regional and developmental classifications. The degree of inter and intra spillover among developed and developing nations is significantly higher than that of frontier and standalone markets. A regional comparison indicates that the pairwise association of the Americas and Europe is as transmitters, and that of the Middle East and Asian Pacific markets is as recipients of spillover. Regardless of the regions and their roles as transmitter or receivers, the overall trend is directly and significantly associated with the degree of market development. Overall, the time-varying pairwise assessment concurs with the individual results' average and time-varying dynamics. Thereby establishing the significance of considering geographical and developmental demographics for portfolio diversification.

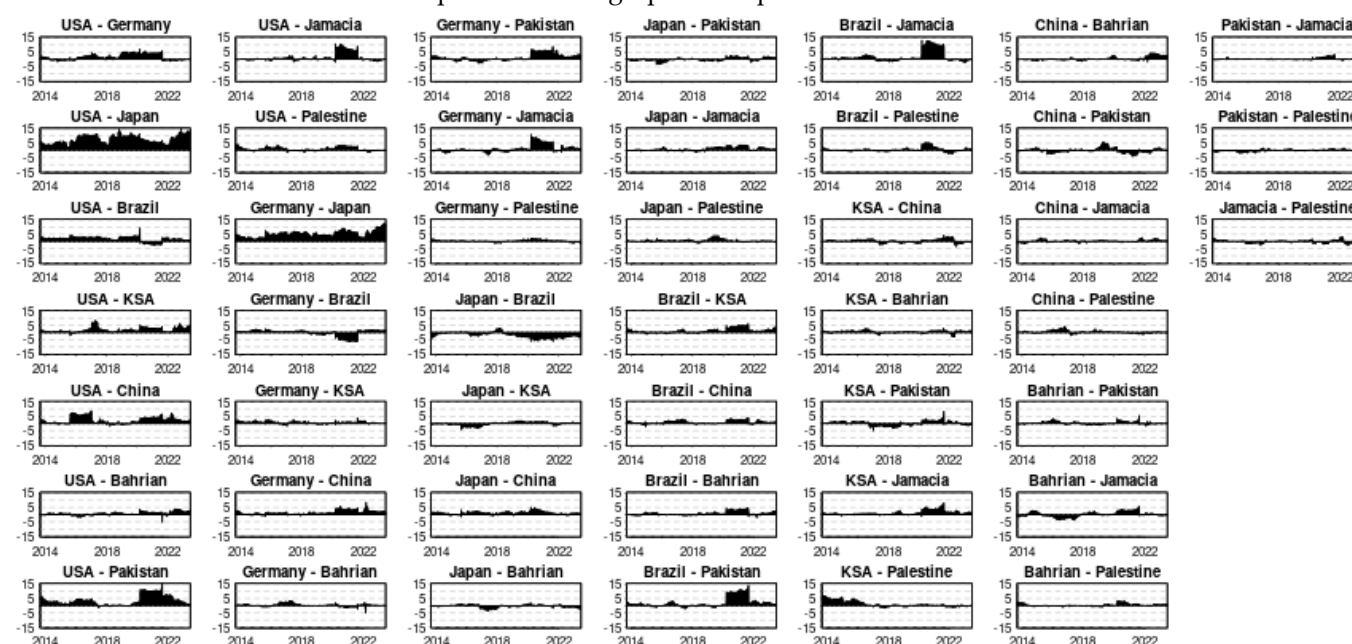


Figure 4-Net Pairwise Spillover

Hedging Evaluation

The varying average, time varying and pairwise spillover results enabled the study to establish the model's validity for hedging effectiveness. Therefore, the findings of the

hedging model are reported in this section. Table 7 represents the pairwise evaluation of the hedging potential among 90 combinations of these stock indices. The findings comprise the hedge ratio (HR), optimal portfolio weights (PW), and hedging effectiveness (HE). The section also compared the effectiveness of two portfolio-building approaches, minimum variance and minimum correlation portfolio. Except for the pairwise association between the USA and Brazil (US/BR), every other portfolio demonstrated a significant p-value ($p < 0.05$) for diversification. Further affirmation can be observed through the zero value of hedging effectiveness (HE). The findings of the USA and Brazil can also be justified through the spillover findings and the study of Broadstock et al. (2022). Since both markets are in the Americas, the higher end of market development caused higher spillover, leading to hedging ineffectiveness. Apart from that, all other portfolios' combinations revealed significant diversification and risk reduction. In contrast, both portfolios' combinations of China with Japan (CN/JP, JP/CN) and China with Bahrain (CN/BH, BH/CN) produced negative Sharpe ratios of -0.06 and -0.11, respectively. These findings indicate underperformance as compared to their benchmark index. These findings can also be justified through extending the interpretation provided by Broadstock et al. (2022). Given that all these markets are in the same region, the connectedness is expected to be much higher, leading towards ineffective hedging capabilities. Only 05 out of 90 possible combinations revealed insignificant or a lack of hedging potential.

Table 5- Pairwise Portfolio Results

Pairs	PW	HE	HR	Pairs	PW	HE	HR
US/DE	0.710	0.130	0.46	CN/US	0.360	0.630	0.13
US/JP	0.660	0.420	0.11	CN/DE	0.470	0.500	0.12
US/BR	0.820	0.000	0.28	CN/JP	0.520	0.410	0.29
US/SA	0.500	0.500	0.13	CN/BR	0.610	0.410	0.09
US/CN	0.640	0.460	0.07	CN/SA	0.360	0.610	0.15
US/BH	0.280	0.830	0.08	CN/BH	0.180	0.870	0.09
US/PK	0.570	0.520	0.04	CN/PK	0.430	0.620	0.11
US/JM	0.490	0.640	-0.03	CN/JM	0.350	0.690	0.05
US/PS	0.220	0.870	0.10	CN/PS	0.130	0.890	0.16
DE/US	0.290	0.330	0.72	BH/US	0.720	0.210	0.03
DE/JP	0.560	0.390	0.24	BH/DE	0.810	0.070	0.03
DE/BR	0.670	0.180	0.23	BH/JP	0.850	0.120	0.04
DE/SA	0.380	0.500	0.25	BH/BR	0.890	0.100	0.03
DE/CN	0.530	0.430	0.11	BH/SA	0.750	0.140	0.11
DE/BH	0.190	0.840	0.12	BH/CN	0.820	0.150	0.02
DE/PK	0.460	0.550	0.12	BH/PK	0.790	0.160	0.02
DE/JM	0.390	0.660	-0.02	BH/JM	0.720	0.240	0.00
DE/PS	0.150	0.890	0.14	BH/PS	0.420	0.580	0.03
JP/US	0.340	0.580	0.21	PK/US	0.430	0.520	0.06
JP/DE	0.440	0.420	0.29	PK/DE	0.540	0.420	0.10
JP/BR	0.600	0.370	0.10	PK/JP	0.590	0.420	0.13
JP/SA	0.340	0.580	0.26	PK/BR	0.670	0.330	0.03
JP/CN	0.480	0.380	0.31	PK/SA	0.430	0.520	0.11
JP/BH	0.150	0.860	0.20	PK/CN	0.570	0.450	0.08

JP/PK	0.410	0.570	0.19	PK/BH	0.210	0.810	0.09
JP/JM	0.340	0.680	0.00	PK/JM	0.420	0.610	-0.01
JP/PS	0.110	0.880	0.22	PK/PS	0.160	0.850	0.14
BR/US	0.180	0.550	0.76	JM/US	0.510	0.430	-0.03
BR/DE	0.330	0.520	0.39	JM/DE	0.610	0.320	-0.01
BR/JP	0.400	0.610	0.14	JM/JP	0.660	0.310	0.00
BR/SA	0.260	0.700	0.23	JM/BR	0.740	0.230	0.00
BR/CN	0.390	0.610	0.14	JM/SA	0.510	0.450	-0.01
BR/BH	0.110	0.910	0.22	JM/CN	0.650	0.310	0.03
BR/PK	0.330	0.690	0.06	JM/BH	0.280	0.740	-0.01
BR/JM	0.260	0.770	-0.01	JM/PK	0.580	0.400	-0.01
BR/PS	0.100	0.930	-0.08	JM/PS	0.220	0.800	-0.01
SA/US	0.500	0.460	0.14	PS/US	0.780	0.240	0.02
SA/DE	0.620	0.300	0.16	PS/DE	0.850	0.190	0.02
SA/JP	0.660	0.370	0.14	PS/JP	0.890	0.100	0.03
SA/BR	0.740	0.280	0.09	PS/BR	0.900	0.120	-0.01
SA/CN	0.640	0.380	0.09	PS/SA	0.780	0.280	0.02
SA/BH	0.250	0.790	0.34	PS/CN	0.870	0.140	0.02
SA/PK	0.570	0.480	0.08	PS/BH	0.580	0.480	0.02
SA/JM	0.490	0.610	-0.01	PS/PK	0.840	0.170	0.03
SA/PS	0.220	0.860	0.07	PS/JM	0.780	0.270	0.00

The Sharpe Ratios (SR) evaluation for these pairwise combinations also revealed interesting insights and ranges from 0.00 to 1.39. The SR's findings supported the study's regional and developmental rationale. Now, we identify and discuss the best and least diversified combination in the sampled data. The findings established that the top 3 best diversified portfolios include Brazil with Palestine (BR/PS), Brazil with Bahrain (BR/BH), and Germany with Palestine (DE/PS), having hedging effectiveness of 93%, 91% and 89%, respectively. To achieve optimal results for each pair an investment of an investment of 10% of total portfolio investment need to be invested in Brazilian market index for BR/PS, 10% of total portfolio investment need to be invested in Brazilian market index for BR/BH, and 15% of total portfolio investment need to be invested in German market index for DE/PS pair. Meanwhile, the findings established that the top 3 worst diversified portfolios among sampled markets include USA with Brazil (US/BR), Bahrain with Germany (BH/DE), and Bahrain with Brazil (BH/BR) having hedging effectiveness of 00%, 07% and 10%, respectively.

Similarly, an overview of the hedge ratios reveals some interesting patterns. The values to create a hedge become as expensive as 0.76 whereas the cheapest values are as low as 0.02. Interestingly, hedge ratios across 12 pairs exuberated negative values. These negative values are indicative of the fact that a dollar long position in base index can't be hedged through a short position in that index (US/JM, DE/JM, BR/JM, BR/PS, SA/JM, PK/JM, JM/US, JM/GR, JM/SA, JM/BH, JM/PK, & JM/PS). Instead, they can diversify by taking similar positions across the markets. In contrast, the hedging ratio 00 is associated with 05 pairs (JP/JM, BH/JM, JM/JP, JM/BR, & PS/JM). These pairs indicate an absence of hedging potential for the base markets by taking the opposing position in their counterpart index. There is an obvious pattern in these pairs, and it can be observed that the Jamaican stock market index pairs failed to provide any hedging benefits to the other

market indices. Jamaica is a standalone market in the American continent. Therefore, it can be attributed to the American region's higher spillover and connectedness to the other global markets. To further support this inference, an analysis of the most expensive hedges highlights that the top 3 most expensive pairs comprise BZ/US, DE/US, and US/DE with values of 0.76, 0.72 and 0.49, respectively. These values can be interpreted as a short position of 76 cents in the Brazilian market index with a long position of 24 cents in the USA market index, which can provide an effective hedge to investors. At the same time, a short position of 72 cents in the USA market index with a long position of 26 cents in the USA market index and a short position of 49 cents in the USA market index with a long position of 51 cents in the German market index are found to create an effective hedge. These expensive and ineffective hedging results indicate a clear pattern of ineffectiveness for the American region. Hence, these findings are aligned with the findings of Broadstock et al. (2022). Now, we come to the seven cheapest hedging combinations among the sampled market indices. The combinations between BH/CN, BH/PK, PS/US, PS/DE, PS/SA, PS/CN, and PS/BH reported similar results with hedging ratio of 0.02. These findings reveal that merely a 2 cents short position with a 98 cents long position in the base markets can provide effective hedge against the portfolio risks.

These findings uncovered that the hedging potential of inter and intra pairs of American and European markets is significantly lower as compared to their combinations with Middle Eastern and Asian Pacific markets. Whereas for the association of Americas and Europe is higher for Middle Eastern market when compared with Asian Pacific region. The Asian Pacific region is found to be the most isolated region in terms of spillover and hedging. The intra connectedness for each region is found to be higher for each region hence establishing the validity of considering regional factors for portfolio diversification, hedging and risk reduction. Lastly, the hedging effectiveness of portfolio pairs varies significantly depending on the difference in their level of market development. These findings are aligned with the findings of Broadstock et al. (2022) regarding regional variations, while also aligning with Tan et al., (2024) and Thaler (2015) suggestion of considering macroeconomic behavioural variations for financial decision making.

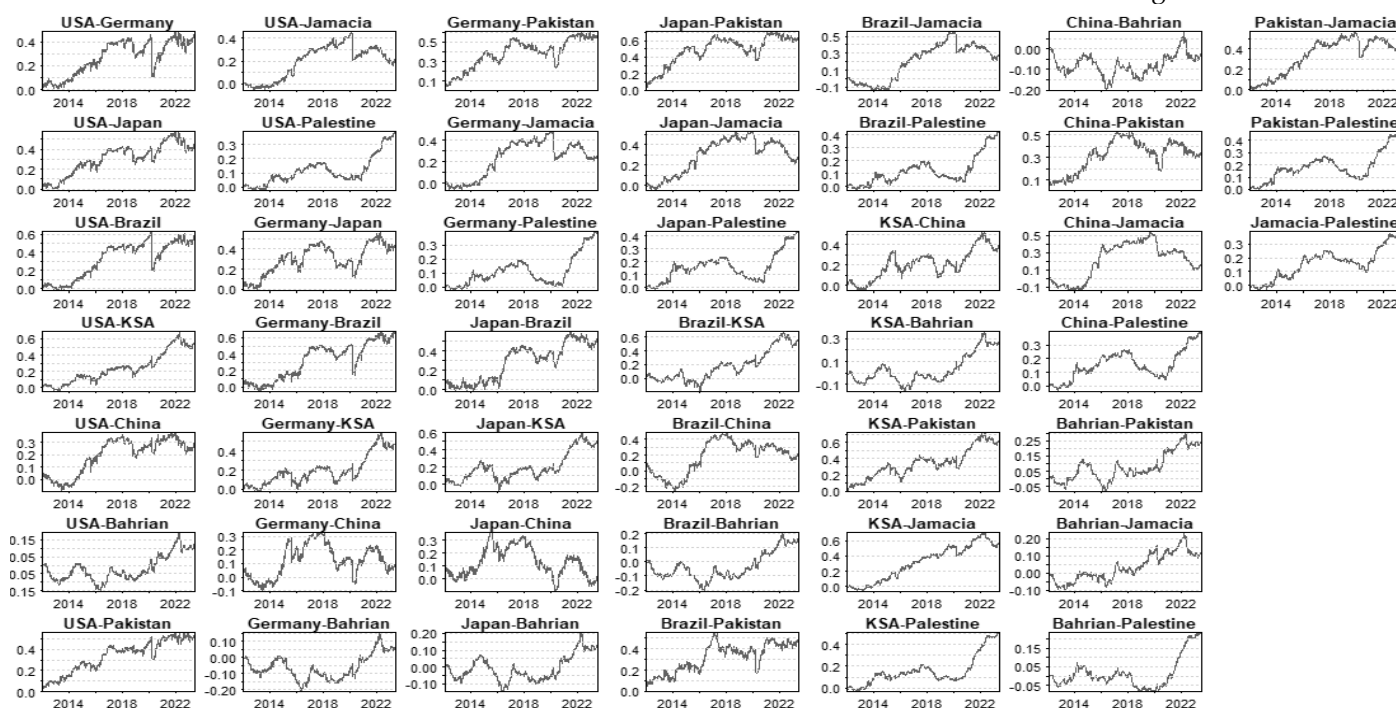


Figure 5- Minimum Variance Based Pairwise Returns

An analysis of the sampled indices using a minimum variance-based portfolio (MVP) and a minimum correlation-based portfolio (MCP) is conducted to evaluate these factors' relevance further. Figure 5 and Figure 6 present the pairwise dynamics of portfolio return for the minimum variance and minimum correlation portfolios. The time-varying analysis will reveal detailed dynamics and abilities of the portfolios, which often remain overlooked in the average results.

On inspection of the minimum variance results it can be observed that apart from the volatility and a declined return associated COVID-19 pandemic period, some other portfolios like DE/CN, DE/BH, JP/BH, BR/CN, BR/BH, BR/JM, SA/BH, CN/BH, CN/JM, BH/PK, BH/JM, BH/PS observed negative returns starting from 2014. Given that most of these pairs are associated with the Bahraini economy, a deeper exploration indicates a sudden decline in oil prices during that tenure aided with the Russian conflict with Ukraine, and an economic slowdown in China has also initiated an adjustment in the Bahrain stock market index. Overall, the return of the minimum variance portfolio is on the rise and remained consistent throughout the sampled period, validating the study's statistical findings. Meanwhile, the time-varying findings of minimum correlation portfolios revealed significantly higher returns while exuberating similar behaviour across the timeline of portfolio pairs.

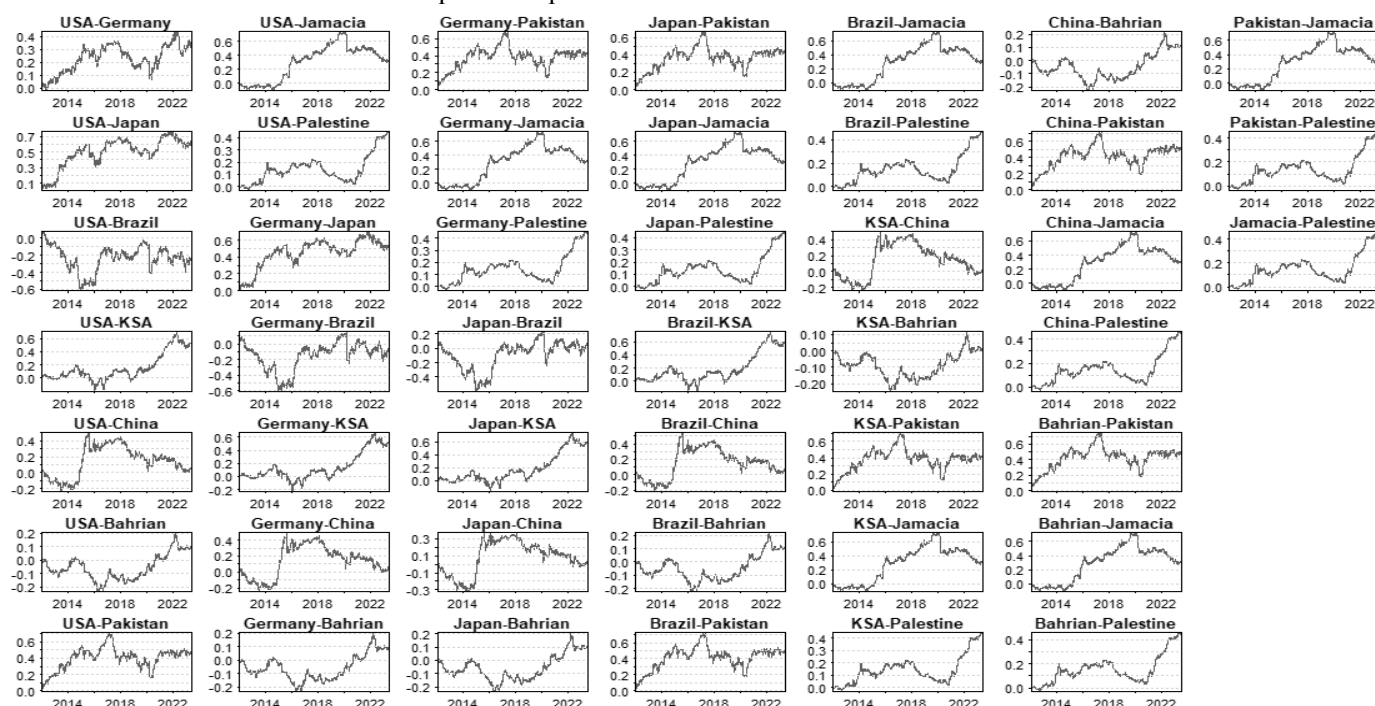


Figure 6—Minimum Correlation Based Pairwise Returns

Table 8 presents the average portfolio results based on the minimum variance approach of Markowitz (1952). The overall Sharpe ratio of the portfolio is 1.05, indicating significant overperformance compared to their base stock indices. Meanwhile, all the results are statistically significant at $p < 0.01$. Furthermore, an average portfolio with 09%, 01%, 01%, 01%, 06%, 03, 26%, 05%, 11%, and 37% weight across US, DE, JP, BR, SA, CN, BH, PK, JM and PS can effectively reduce volatility in the portfolio by 93%, 95%, 95%, 97%, 93%, 95%, 70%, 93%, 90% and 63% respectively.

Table 6- Minimum Variance Portfolios

	Mean	Std.Dev.	5%	95%	HE	p-value
US	0.090	0.070	0.000	0.240	0.930	0.000
DE	0.010	0.020	0.000	0.050	0.950	0.000
JP	0.010	0.010	0.000	0.030	0.950	0.000
BR	0.010	0.010	0.000	0.030	0.970	0.000
SA	0.060	0.060	0.000	0.180	0.930	0.000
CN	0.030	0.020	0.000	0.080	0.950	0.000
BH	0.260	0.110	0.090	0.450	0.700	0.000
PK	0.050	0.030	0.010	0.100	0.930	0.000
JM	0.110	0.030	0.060	0.160	0.900	0.000
PS	0.370	0.110	0.180	0.540	0.630	0.000

Similar to the findings of the minimum variance portfolio, Table 9 presents the average portfolio results based on the minimum correlation approach of Christoffersen et al. (2014). The correlation-based portfolios yield significantly higher returns in the pairwise analysis of the current study. Here, the overall Sharpe ratio of the portfolio is 0.93, indicating significant overperformance compared to its base stock indices. Meanwhile, all the results are statistically significant at $p < 0.01$. Furthermore, an average portfolio with 07%, 04%, 06%, 09%, 08%, 09%, 13%, 12%, 17% and 14% weight across US, DE, JP, BR, SA, CN, BH, PK, JM and PS can effectively reduce volatility in the portfolio by 83%, 87%, 88%, 92%, 82%, 89%, 25%, 83%, 74%, and 07% respectively.

Table 7- Minimum Correlation Portfolios

	Mean	Std.Dev.	5%	95%	HE	p-value
US	0.070	0.010	0.060	0.080	0.830	0.000
DE	0.040	0.010	0.030	0.050	0.870	0.000
JP	0.060	0.000	0.050	0.070	0.880	0.000
BR	0.090	0.010	0.080	0.100	0.920	0.000
SA	0.080	0.000	0.080	0.090	0.820	0.000
CN	0.090	0.000	0.090	0.100	0.890	0.000
BH	0.130	0.000	0.120	0.130	0.250	0.000
PK	0.120	0.000	0.120	0.130	0.830	0.000
JM	0.170	0.000	0.170	0.180	0.740	0.000
PS	0.140	0.000	0.140	0.150	0.070	0.160

Overall, the findings of the minimum variance portfolio (MVP) and minimum correlation portfolio (MCP) provide the robustness of the earlier findings regarding the hedging potential of the regional and developmental classification of MSCI-based indices. Both portfolios significantly reduced portfolio volatility in their conclusions. Hence, it validates the potential of each stock market index for increased diversification and hedging against market volatility. Even though these portfolios were constructed without considering the regional dynamics and variations, they still produced results consistent with the study's rationale. After considering the high degree of spillover among American and European markets, a portfolio reconstruction can result in even more diversified and optimal portfolios.

Lastly, Figure 7 provides insights into the spatial variation of time-varying dynamics in the portfolio weights of MVP (Black) and MCP(Blue). The finding uncovers that although the overall trend of the majority index weights across sampled time remains consistent for minimum variance portfolio, significant fluctuations can be observed across the USA, Bahrain and Palestine. Despite their time-varying nature, no abrupt changes can be observed from these time-varying weights, but overall, a consistent behaviour is observable. Meanwhile, the time-varying weights of the minimum correlation portfolio present much more stable and non-volatile behaviours across the timeline. These findings indicate that the minimum correlation portfolio remained consistent even during financial distress, uncertainty and decline.

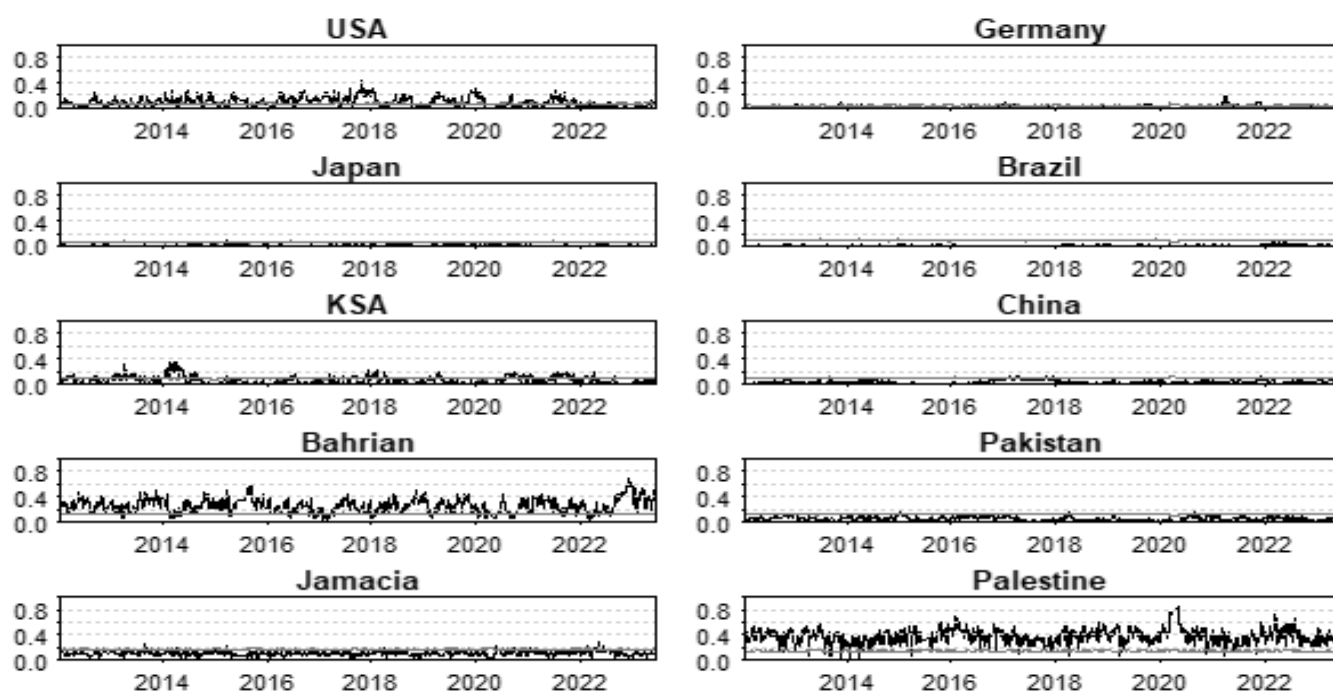


Figure 7-Portfolio Weights

4. Conclusions

The current investigation adopted a behavioral explanation of stock markets behavior to evaluate the hedging potential of demographic distinctions. The inception of behavioral finance sparked a wave of contextualized understanding of micro and macro determinants of financial behavior. Contrary to the traditional financial models' behavioral models posits that economic behavior varies significantly across individual and economies. The recent rise in technology further elevated these discrepancies and deviations while also heightening the degree of connectivity across the globe, creating a highly volatile and delicate global financial environment. The increasing international economic integration makes it difficult for investors to diversify their portfolio effectively to gain optimal returns. This has created a wave among researchers to uncover and explore unique combinations of portfolios. The current study is a pioneer in studying stock market spillover and portfolio efficiency through the lens of behavioral distinctions of stock markets.

The current study utilized data from ten representative stock indices from the MSCI classification of stock market development and regions. The stock indices return dataset ranges from 2012 to 2023, and a VAR-based connectedness approach was used to test for the degree of spillover among these markets. The findings revealed a moderate degree of connectedness across stock markets, which diminished with decreased financial development. The developed economies have the highest level of spillovers across each other.

Furthermore, the findings established that interregional spillovers are significantly lower than the intraregional connectedness among stock indices. Meanwhile, the connectedness between the Americas, Europe, and the Middle East & Africa region is higher than with the Asia-Pacific region. The time-varying analysis of the connectedness model further reveals that connectedness became significantly higher during the COVID-19 pandemic. Lastly, to uncover hedging potential, a GARCH-based methodology is applied. The findings of the minimum variance portfolio (MVP) and minimum correlation portfolio (MCP) established that investors can effectively hedge their portfolios by considering regional and developmental differences across the globe. Hence behavioral distinctions are found to play a distinct role in creating value for portfolio holders meanwhile significantly reducing the risks associated with their investments.

The study produces significant contributions on multiple avenues including behavioral finance, regional finance, market connectedness and hedging theory. Theoretically the study expanded the current understanding and application of behavioral models into the realm of creating effective hedging strategies. Meanwhile, the results of the study are also relevant to the diverse array of audience. For investing stakeholders, the study provides the relevance and significance of considering regional and developmental differences to achieve higher portfolio performance. Furthermore, the knowledge gained from the finding provides critical insights to the issuers of index funds and it illustrates that index funds can further be diversified by considering other index funds in their portfolios. Lastly, the exploration of this unique avenue is extremely relevant for policy makers and regulators of the stock markets. On one end the study provides an overview of the stock market while on other hand, the findings of the study explain the movement and sensitivity of the regional and developmental levels. This better equip the stakeholder to create more effective and efficient market frameworks to create stabile and sustainable financial environment.

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Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronyms
LD	Linear dichroism

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